

Chapter 5. Eigenvalues and Eigenvectors

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5.1. Eigenvectors and Eigenvalues

Eigenvectors and Eigenvalues

● Definition

- An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ .
- A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$, such an $\mathbf{x}$ is called an *eigenvector corresponding to λ* .

● Remark

- λ is an eigenvalue of an $n \times n$ matrix A if and only if the following equation has nontrivial solution.

$$(A - \lambda I)\mathbf{x} = 0 \tag{3}$$

- The set of all solutions of (3) is just the null space of the matrix $A - \lambda I$.
- So this set is a subspace of $\mathbb{R}^n$ and is called the **eigenspace** of A corresponding to λ .
- The eigenspace consists of the zero vector and all the eigenvectors corresponding to λ .

- **Example 3:** Show that 7 is an eigenvalue of matrix $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ and find the corresponding eigenvectors.

- **Solution:** (proof that 7 is eigenvalue)

- The scalar 7 is an eigenvalue of A if and only if the equation has a nontrivial solution.

$$A\mathbf{x} = 7\mathbf{x} \quad (1)$$

- But (1) is equivalent to $A\mathbf{x} - 7\mathbf{x} = 0$ or

$$(A - 7I)\mathbf{x} = 0 \quad (2)$$

- To solve this homogeneous equation, form the matrix

$$A - 7I = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix}$$

- The columns of $A - 7I$ are obviously linearly dependent, so (2) has nontrivial solutions.

● **Solution:** (finding its corresponding eigenvector)

- To find the corresponding eigenvectors, use row operations:

$$\begin{bmatrix} -6 & 6 & 0 \\ 5 & -5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- The general solution has the form $x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
- Each vector of this form with $x_2 \neq 0$ is an eigenvector corresponding to $\lambda = 7$.
- (If you multiply a constant to an eigenvector, it is again an eigenvector.)

- **Example 4:** Let $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$. An eigenvalue of A is 2. Find a basis for the corresponding eigenspace.

- **Solution:**

- Form

$$A - 2I = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

and row reduce the augmented matrix for $(A - 2I)\mathbf{x} = 0$.

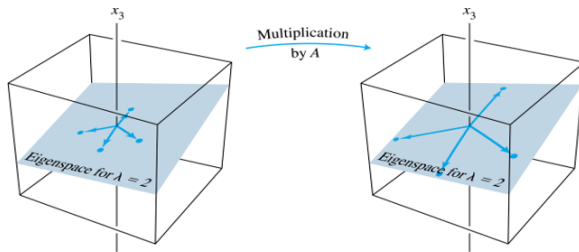
$$\begin{bmatrix} 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- At this point, it is clear that 2 is indeed an eigenvalue of A because the equation $(A - 2I)\mathbf{x} = 0$ has free variables.
- The general solution is $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$, x_2 and x_3 free.

• (Solution continued)

- The eigenspace, shown in the following figure, is a two-dimensional subspace of $\mathbb{R}^3$. A basis is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$



A acts as a dilation on the eigenspace.

- **Theorem 1:** The eigenvalues of a triangular matrix are the entries on its main diagonal.

- **Proof:**

- For simplicity, consider the 3×3 case. If A is upper triangular, the $A - \lambda I$ has the form

$$A - \lambda I = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ 0 & a_{22} - \lambda & a_{23} \\ 0 & 0 & a_{33} - \lambda \end{bmatrix}$$

- The scalar λ is an eigenvalue of A if and only if the equation $(A - \lambda I)\mathbf{x} = 0$ has a nontrivial solution, that is, if and only if the equation has a free variable.
- Because of the zero entries in $A - \lambda I$, it is easy to see that $(A - \lambda I)\mathbf{x} = 0$ has a free variable if and only if at least one of the entries on the diagonal of $A - \lambda I$ is zero.
- This happens if and only if λ equals one of the entries a_{11}, a_{22}, a_{33} in A .
- Sim: “The equation $((A - \lambda I)\mathbf{x} = 0)$ ” has a nontrivial solution if and only if $|A - \lambda I| = 0$.
- Sim: This equation is also called as *characteristic polynomial*.

- **Theorem 2:** If $\mathbf{v}_1, \dots, \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of an $n \times n$ matrix A , then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent.

- **Proof:**

- Suppose $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly dependent. Since $\mathbf{v}_1$ is nonzero, Theorem 7 in Section 1.7 says that one of the vectors in the set is a linear combination of the preceding vectors.
- Let p be the least index such that $\mathbf{v}_{p+1}$ is a linear combination of the preceding (linearly independent) vectors.
- Then, there exist scalars $c_1, \dots, c_p$ such that

$$c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p = \mathbf{v}_{p+1} \quad (5)$$

- Multiplying both sides of (5) by A and using the fact that $A\mathbf{v}_k = \lambda_k \mathbf{v}_k$ for each k , we obtain $c_1 A\mathbf{v}_1 + \dots + c_p A\mathbf{v}_p = A\mathbf{v}_{p+1}$

$$c_1 \lambda_1 \mathbf{v}_1 + \dots + c_p \lambda_p \mathbf{v}_p = \lambda_{p+1} \mathbf{v}_{p+1} \quad (6)$$

- Multiplying both sides of (5) by λ_{p+1} and subtracting the result from (6), we have

$$c_1 (\lambda_1 - \lambda_{p+1}) \mathbf{v}_1 + \dots + c_p (\lambda_p - \lambda_{p+1}) \mathbf{v}_p = 0 \quad (7)$$

- Since $\mathbf{v}_1, \dots, \mathbf{v}_r$ is linearly independent, the weights in (6) are all zero.

● (Proof continued)

- But none of the factors $\lambda_i - \lambda_{p+1}$ are zero, because the eigenvalues are distinct. Hence, $c_i = 0$ for $i = 1, \dots, p$.
- But then (4) says that $\mathbf{v}_{p+1} = 0$, which is impossible.
- Hence $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ cannot be linearly dependent and therefore must be linearly independent.
- If A is an $n \times n$ matrix, then (8) is a recursive description of a sequence $\{x_k\}$ in $\mathbb{R}^n$.

$$\mathbf{x}_{k+1} = A\mathbf{x}_k \quad (k = 0, 1, 2, \dots) \quad (8)$$

- A **solution** of (8) is an explicit description of $\{x_k\}$ whose formula for each x_k does not depend directly on A or on the preceding terms in the sequence other than the initial term $\mathbf{x}_0$.
- The simplest way to build a solution of (8) is to take an eigenvector $\mathbf{x}_0$ and its corresponding eigenvalue λ and let

$$\mathbf{x}_k = \lambda^k \mathbf{x}_0 \quad (k = 1, 2, \dots) \quad (9)$$

- This sequence is a solution because

$$A\mathbf{x}_k = A(\lambda^k \mathbf{x}_0) = \lambda^k (A\mathbf{x}_0) = \lambda^k (\lambda \mathbf{x}_0) = \lambda^{k+1} \mathbf{x}_0 = \mathbf{x}_{k+1}$$

□

Suggested Exercises

- 5.1.3
- 5.1.6
- 5.1.15

5.2. The Characteristic Equation

Determinants (review of)

- Let A be an $n \times n$ matrix, let U be any echelon form obtained from A by row replacements and row interchanges (without scaling), and let r be the number of such row interchanges.
- Then the **determinant** of A , written as $\det A$, is $(-1)^r$ times the product of the diagonal entries $u_{11}, \dots, u_{nn}$ in U .
- If A is invertible, then $u_{11}, \dots, u_{nn}$ are all pivots (because $A \sim I_n$ and the u_{ii} have not been scaled to 1's).
- Otherwise, at least u_{nn} is zero, and the product $u_{11}, \dots, u_{nn}$ is zero.
- Thus,

$$\det A = \begin{cases} (-1)^r \cdot (\text{product of pivots in } U), & \text{when } A \text{ is invertible} \\ 0, & \text{when } A \text{ is not invertible} \end{cases}$$

- **Example 1:** Compute $\det A$ for $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$.

- **Solution:**

- The following row reduction uses one row interchange:

$$A \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} = U_1$$

- So $\det A$ equals $(-1)^1(1)(-2)(-1) = -2$.
- The following alternative row reduction avoids the row interchange and produces a different echelon form.
- The last step adds $-1/3$ times row 2 to row 3:

$$A \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 1/3 \end{bmatrix} = U_2$$

- This time $\det A$ is $(-1)^0(1)(-6)(1/3) = -2$, the same as before.

Theorems

- **The invertible matrix theorem (continued):** Let A be an $n \times n$ matrix. Then A is invertible if and only if:
 - s. The number 0 is not an eigenvalue of A .
 - t. The determinant of A is *not* zero.

- **Theorem 3: (Properties of Determinants)** Let A and B be $n \times n$ matrices.
 - a. A is invertible if and only if $\det A \neq 0$.
 - b. $\det AB = (\det A)(\det B)$.
 - c. $\det A^T = \det A$.
 - d. If A is triangular, then $\det A$ is the product of the entries on the main diagonal of A .
 - e. A row replacement operation on A does not change the determinant. A row interchange changes the sign of the determinant. A row scaling also scales the determinant by the same scalar factor.

- **Remark**
 - Theorem 3(a) shows how to determine when a matrix of the form $A - \lambda I$ is not invertible.

The characteristic equation

- The scalar equation $\det(A - \lambda I) = 0$ is called the characteristic equation (or, characteristic polynomial) of A .
- A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation

$$\det(A - \lambda I) = 0$$

- **Example 3:** Find the characteristic equation of

$$A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Solution:**

- Form $A - \lambda I$, and use Theorem 3(d):

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} 5 - \lambda & -2 & 6 & -1 \\ 0 & 3 - \lambda & -8 & 0 \\ 0 & 0 & 5 - \lambda & 4 \\ 0 & 0 & 0 & 1 - \lambda \end{bmatrix} \\ &= (5 - \lambda)(3 - \lambda)(5 - \lambda)(1 - \lambda) \end{aligned}$$

- The characteristic equation is

$$(5 - \lambda)^2(3 - \lambda)(1 - \lambda) = 0 \text{ or } (\lambda - 5)^2(\lambda - 3)(\lambda - 1) = 0$$

- Expanding the product, we can also write

$$\lambda^4 - 14\lambda^3 + 68\lambda^2 - 130\lambda + 75 = 0$$

● Remark

- If A is an $n \times n$ matrix, then $\det(A - \lambda I)$ is a polynomial of degree n called the **characteristic polynomial** of A .
- The eigenvalue 5 in Example 3 is said to have *multiplicity* 2 because $(\lambda - 5)$ occurs two times as a factor of the characteristic polynomial.
- In general, the **(algebraic) multiplicity** of an eigenvalue λ is its multiplicity as a root of the characteristic equation.

Similarity

- If A and B are $n \times n$ matrices, then A is **similar to** B if there is an invertible matrix P such that $P^{-1}AP = B$, or, equivalently, $A = PBP^{-1}$.
- Writing Q for P^{-1} , we have $Q^{-1}BQ = A$.
- So B is also similar to A , and we say simply that A and B **are similar**.
- Changing A into $P^{-1}AP$ is called a **similarity transformation**.

- **Theorem 4:** If $n \times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

- **Proof:**

- If $B = P^{-1}AP$ then,
$$B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(AP - \lambda P) = P^{-1}(A - \lambda I)P$$
- Using the multiplicative property in Theorem 3(b), we compute
$$\det(B - \lambda I) = \det[P^{-1}(A - \lambda I)P] = \det(P^{-1}) \cdot \det(A - \lambda I) \cdot \det(P)$$
- Since $\det(P^{-1}) \cdot \det(P) = \det(P^{-1}P) = \det I = 1$, we see from the previous equation that $\det(B - \lambda I) = \det(A - \lambda I)$.

● Warnings:

1. The matrices are not similar even though they have the same eigenvalues.

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

2. Similarity is not the same as row equivalence. (If A is row equivalent to B , then $B = EA$ for some invertible matrix E). Row operations on a matrix usually change its eigenvalues.

Suggested Exercises

- 5.2.3

5.3. Diagonalization

Diagonalization (its benefit and definition)

- **Example 2:** Let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a formula for A^k , given that $A = PDP^{-1}$, where $P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$

- **Solution:**

- The standard formula for the inverse of a 2×2 matrix yields $P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$
- Then, by associativity of matrix multiplication,

$$\begin{aligned} A^2 &= (PDP^{-1})(PDP^{-1}) = PD \underbrace{(P^{-1}P)}_I DP^{-1} = PDDP^{-1} \\ &= PD^2P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} \end{aligned}$$

- Again,

$$A^3 = (PDP^{-1})A^2 = (PD \underbrace{P^{-1}P}_I) D^2 P^{-1} = PDD^2P^{-1} = PD^3P^{-1}$$

● (solution continued)

- In general, for $k \geq 1$,

$$\begin{aligned} A^k &= PD^kP^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \cdot 5^k - 3^k & 5^k - 3^k \\ 2 \cdot 3^k - 2 \cdot 5^k & 2 \cdot 3^k - 5^k \end{bmatrix} \end{aligned}$$

The diagonalization theorem

- **Definition:** A square matrix A is said to be **diagonalizable** if A is similar to a diagonal matrix, that is, if $A = PDP^{-1}$ for some invertible matrix P and some diagonal, matrix D .
- **Theorem 5 (diagonalization theorem):**
 - a. An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.
 - b. In other words, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A .
 - c. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .
- **Remark:** In other words, A is diagonalizable if and only if there are enough eigenvectors to form a basis of $\mathbb{R}^n$. We call such a basis an **eigenvector basis** of $\mathbb{R}^n$.

● **Proof** (only if part of *a* & *b*; and the statement *c*)

- First, observe that if P is any $n \times n$ matrix with columns $\mathbf{v}_1, \dots, \mathbf{v}_n$, and if D is any diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then

$$AP = A[\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n] = [A\mathbf{v}_1 \ A\mathbf{v}_2 \ \cdots \ A\mathbf{v}_n] \quad (1)$$

while

$$PD = P \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} = [\lambda_1 \mathbf{v}_1 \ \lambda_2 \mathbf{v}_2 \ \cdots \ \lambda_n \mathbf{v}_n] \quad (2)$$

- Now suppose A is diagonalizable and $A = PDP^{-1}$. Then right-multiplying this relation by P , we have $AP = PD$.
- In this case, equations (1) and (2) imply that

$$[A\mathbf{v}_1 \ A\mathbf{v}_2 \ \cdots \ A\mathbf{v}_n] = [\lambda_1 \mathbf{v}_1 \ \lambda_2 \mathbf{v}_2 \ \cdots \ \lambda_n \mathbf{v}_n] \quad (3)$$

- Equating columns, we find that

$$A\mathbf{v}_1 = \lambda_1 \mathbf{v}_1, A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2, \dots, A\mathbf{v}_n = \lambda_n \mathbf{v}_n \quad (4)$$

● (proof continued)

- Since P is invertible, its columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ must be linearly independent.
- Also, since these columns are nonzero, the equations in (4) show that $\lambda_1, \dots, \lambda_n$ are eigenvalues and $\mathbf{v}_1, \dots, \mathbf{v}_n$ are corresponding eigenvectors.
- This argument proves the “only if” parts of the first and second statements, along with the third statement, of the theorem.

● **Proof** (if part of *a* & *b*)

- Finally, given any n eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, use them to construct the columns of P and use corresponding eigenvalues $\lambda_1, \dots, \lambda_n$ to construct D .
- By equations (1)–(3), $AP = PD$.
- This is true without any condition on the eigenvectors.
- If, in fact, the eigenvectors are linearly independent, then P is invertible (by the Invertible Matrix Theorem), and $AP = PD$ implies that $A = PDP^{-1}$. $\square$

Diagonalizing Matrices

- **Example 3:** Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

- **Solution:** There are four steps to implement the description in Theorem 5.
 - **Step 1. Find the eigenvalues of A .**
 - Here, the characteristic equation turns out to involve a cubic polynomial that can be factored: $0 = \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4 = -(\lambda - 1)(\lambda + 2)^2$
 - The eigenvalues are $\lambda = 1$ and $\lambda = -2$.
 - **Step 2. Find three linearly independent eigenvectors of A .**
 - Three vectors are needed because A is a 3×3 matrix.
 - This is a critical step. If it fails, then Theorem 5 says that A cannot be diagonalized.

● (solution continued)

- Basis for $\lambda = 1$: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$
- Basis for $\lambda = -2$: $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$
- You can check that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set.
- **Step 3. Construct P from the vectors in step 2.**
- The order of the vectors is unimportant. Using the order chosen in step 2, form

$$P = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3] = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

- **Step 4. Construct D from the corresponding eigenvalues.**
- In this step, it is essential that the order of the eigenvalues matches the order chosen for the columns of P .
- Use the eigenvalue $\lambda = -2$ twice, once for each of the eigenvectors corresponding to

$$\lambda = -2: D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

● (solution continued - sanity check)

- To avoid computing P^{-1} , simply verify that $AP = PD$.
- Compute

$$AP = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

$$PD = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

● **Theorem 6:** An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

● **Proof:**

- Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be eigenvectors corresponding to the n distinct eigenvalues of a matrix A .
- Then $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent, by Theorem 2 in Section 5.1.
- Hence A is diagonalizable, by Theorem 5.

Matrices whose eigenvalues are not distinct

● Remarks

- It is not *necessary* for an $n \times n$ matrix to have n distinct eigenvalues in order to be diagonalizable.
- The 3×3 matrix in Example 3 is diagonalizable even though it has only two distinct eigenvalues.
- If an $n \times n$ matrix A has n distinct eigenvalues, with corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, and if $P = [\mathbf{v}_1, \dots, \mathbf{v}_n]$, then P is automatically invertible because its columns are linearly independent, by Theorem 2.
- When A is diagonalizable but has fewer than n distinct eigenvalues, it is still possible to build P in a way that makes P automatically invertible, as the next theorem shows.

Suggested Exercises

- 5.3.12
- 5.3.14
- 5.3.21

Acknowledgement

- This lecture note is based on the instructor's lecture notes (formatted as ppt files) provided by the publisher (Pearson Education) and the textbook authors (David Lay and others)
- The pdf conversion project for this chapter was possible thanks to the hard work by Doyun Kwon (M.S. in Data Science, 21'). Mr. Kwon joined professor Sim's lab mainly due to his interest in unmanned asset management (a.k.a. robo-advisor). His research topic includes devising reinforcement learning-based investment strategy on crypto-currency trading market.
- He earned his Master's degree at Aug 2024 and is working as a researcher in Deep Learning Research Institute of SURROMIND.

